ESHAN KUMAR JAIN

AI/ML Engineer | Generative AI | Deep Learning | NLP | LLM Agents

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PROFESSIONAL SUMMARY

AI/ML engineer with an MSc in Data Science and Analytics (First Class Honours, Maynooth University) and a B.Tech in Computer Science (NIT Nagpur), plus two years of banking technology experience at HDFC Bank. Builds deep learning and generative AI systems end to end: self-supervised transformers and CLIP-style multimodal models in PyTorch, LLM agents driven by engineered multi-step prompt workflows, and statistical and ML models shipped as live FastAPI and Streamlit apps, documented for technical and non-technical stakeholders.

TECHNICAL SKILLS

Generative AI & NLP: LLMs, Transformers, Prompt Engineering, Prompt Chaining, RAG, Vector Databases, Fine-Tuning, LLM Evaluation, AI Agents, Agentic Workflows, LangChain, Hugging Face, OpenAI API, Anthropic Claude API, Model Context Protocol (MCP), Claude Code, CLIP Embeddings, Contrastive Learning, Masked Modelling, NLP

Machine & Deep Learning: Python, PyTorch, TensorFlow, Scikit-learn, XGBoost, Neural Networks, CNNs, Self-Supervised Learning, Classification, Clustering, Anomaly Detection, Feature Engineering, Model Evaluation, SHAP

Statistics, Data & Visualisation: Statistical Modelling, Hypothesis Testing, A/B Testing, Regression, Cohort Analysis, SQL, PostgreSQL, Pandas, NumPy, R, EDA, Data Cleaning, Data Quality Validation, Matplotlib, Seaborn, Power BI, DAX

Cloud & Engineering: Databricks, Vercel, FastAPI, REST APIs, Next.js, TypeScript, Git, Linux, MongoDB, JSON; Data Structures & Algorithms

Delivery: Agile, SDLC, JIRA, Technical Specifications (BRD/FRD), Requirements Elicitation, UAT, Stakeholder Management

WORK EXPERIENCE

AI Data Analyst | Crossing Hurdles | *Remote* **Aug 2026 – Presen**

- Write coding tasks and grading rubrics used to train and evaluate large language models.

Manager, Tech and Digital (Business Data Analyst) | HDFC Bank | *Mumbai* **Jul 2023 – Jul 202**

- Owned the API Integration and Customised Statements workstreams, supporting **INR 15 Cr+** in daily transaction value.
- Automated incident handling with **agentic workflows** that classified incidents, validated payloads and routed issues between JIRA and ServiceNow.
- Cut turnaround time **25%** for 50+ stakeholders by testing bottleneck hypotheses with SQL cohort analysis and Excel trend modelling.
- Reduced data reconciliation errors **30%** by writing PostgreSQL data-quality test plans and leading UAT across 5+ major releases.
- Designed throughput, SLA-adherence and API error-rate dashboards that made management decisions **20%** faster.
- Wrote BRDs/FRDs for a 5+ developer Agile team; owned every EPFO project end to end with full regulatory compliance.

AI / ML PROJECTS

Skin Disease Classification: Lightweight CNN + CLIP Ensemble (Published) | *PyTorch, CLIP, CNNs, Healthcare AI*

- Co-authored a published preprint on 7-class skin disease classification from **10,015** HAM10000 dermoscopy images, averaging two lightweight CNNs (MobileNetV2, ShuffleNetV2) with a CLIP ViT-B/32 linear probe.
- Handled severe class imbalance (one class ~**67%** of images) with balanced class weighting, and judged models on macro F1 so rare but clinically important lesion types counted as much as common ones.
- Ensemble reached macro F1 **0.652** and **82.5%** accuracy on the held-out test set, **8.6 points** above the best single model; caught a VGG19 baseline whose 0.657 validation score hid a complete majority-class collapse.

MSc Thesis: Self-Supervised Learning for Spectral Region Mapping (GitHub) | *PyTorch, Transformers, Contrastive Learning*

- Trained self-supervised models on **96,093** unlabelled VLT/MUSE spectra; a CLIP-style dual encoder reproduced hand-drawn jet boundaries (synthetic **ARI 1.000**).
- Built a masked-modelling transformer fusing two instruments across a 50x resolution gap (**ARI 0.997**).

WealthIQ: Generative AI Stock Analysis Agent (Live) | *Claude API, Prompt Engineering, Next.js, TypeScript, Vercel*

- Deployed an LLM agent on the Claude API with a multi-step prompt workflow that turns NSE/BSE fundamentals into plain-English reports on valuation, growth and financial health.

Financial Data Analysis AI Agent: Journal-Entry Anomaly Detection (Live) | *Python, SQL, Scikit-learn, XGBoost, Streamlit*

- Generated a **114K-entry** synthetic ledger seeded with 8 error types across three difficulty tiers; built SQL feature tables with expanding-window functions.
- Layered statistical models (Benford's Law, segmented z-scores) with Isolation Forest, logistic regression, Random Forest and XGBoost (SHAP); found and fixed a temporal leak and a label leak.

Credit Card Approval Risk Classifier (Live) | *Python, Scikit-learn, XGBoost, FastAPI, Streamlit*

- Spotted that a "91% accurate" model approved every risky applicant (0 of 28 rejections caught); reframed the metric around business risk, lifting **ROC-AUC 0.55 to 0.77**, and shipped it as a human-review triage tool.

EDUCATION

MSc Data Science and Analytics, First Class Honours | Maynooth University, Ireland **Sep 2025 – Aug 20**

B.Tech, Computer Science and Engineering | NIT Nagpur, India **Jul 2019 – Jun 20**

CERTIFICATIONS AND ACHIEVEMENTS

Certifications: Anthropic (Claude 101 Prompt Engineering, Agent Skills, Subagents, MCP), HackerRank SQL, Udemy REST APIs;

Databricks Generative AI Engineer Associate (in preparation)
Job Simulation: Deloitte Australia Data Analytics (Forage) – PostgreSQL JSON parsing, Power BI dashboard, pay-equity analysis
Open Source: pandas, merged PR #69435 (read_sql regression test for SQLAlchemy, GH#35484)
Achievements: JEE Main 99.57 percentile, JEE Advanced AIR 6253, Olympiad Gold Medalist, VP Astronomy Club (NIT Nagpur)